MIL-T-36682A
13 May 1977
SUPERSEDING
MIL-T-36682
11 September 1970

MILITARY SPECIFICATION

TUBE, COILED, BLOOD WARMING

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers one type of disposable, plastic, coiled blood warming tube with a sterile fluid path, suitable for warming bank blood from 35° F to 98° F prior to transfusion.

2. APPLICABLE DOCUMENTS

2.1 Issues of documents. The following documents of the issue in effect on date of invitation for bids or request for proposals form a part of this specification to the extent specified herein:

SPECIFICATIONS

FEDERAL

MMM-A-250	- Adhesive, Water Resistant (for Sealing Fiberboard Boxes).
PPP-B-585	- Boxes, Wood, Wirebound.
PPP-B-601	- Boxes, Wood, Cleated-Plywood.
PPP-B-621	- Boxes, Wood, Nailed and Lock-Corner.
PPP-B-636	- Boxes, Shipping, Fiberboard.
PPP-T-76	- Tape, Pressure-Sensitive Adhesive Paper, (for Carton Sealing).

MILITARY

MIL-S-36586	- Sterilization Strip Test, Bacterial Spore.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Headquarters, Defense Personnel Support Center, ATTN: Directorate of Medical Materiel, DPSC-ATT, 2800 South 20th Street, Philadelphia, PA 19101, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

FSC 6515

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T09:01Z
Check the source to verify that this is the current version before use.

STANDARDS

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129 - Marking for Shipment and Storage.

(Copies of specifications, standards, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposals shall apply:

THE UNITED STATES PHARMACOPEIAL CONVENTION, INC. (USP)

The Pharmacopeia of the United States of America.

(Application for copies shall be addressed to the Mack Publishing Company, Easton, Pennsylvania 18042.)

AMERICAN NATIONAL STANDARDS INSTITUTE (ANSI)

Z70.1 American Standard Dimensions of Glass and Metal Luer Tapers for Medical Applications.

(Application for copies should be addressed to the American National Standards Institute, 1430 Broadway, New York 10018.)

U. S. DEPARTMENT OF HEALTH, EDUCATION AND WELFARE

Federal Food, Drug and Cosmetics Act and General Regulations for its Enforcement.

(Applications for copies should be addressed to the Federal Food, Drug Administration, Washington, D.C. 20204.)

NATIONAL INSTITUTES OF HEALTH

Public Health Service Regulations Part 73 Pertaining to Federal Control of the Sale, Barter or Exchange of Biological Products.

(Application for copies should be addressed to the Laboratory of Biologics Control, National Institutes of Health, Bethesda, Maryland 20014.)

Source: https://assist.dla.mil -- Downloaded: 202610-21T09:01Z
Check the source to verify that this is the current version before use.

3. REQUIREMENTS

3.1 Material. Shall be made of high quality plastics and rubber, suitable in all respects for the intended purpose. Shall be sterile, inert, free from pyrogens, and safe (non-toxic). End protectors and those plastic and rubber components coming in contact with fluids being administered shall contain no more than 5 parts per million (ppm) of lead when tested as specified in 4.4.2. The plastic shall meet the requirements of Transfusion and Infusion Assemblies, and USP Class II Plastics (70° C) in accordance with 4.4.1.

3.1.1 Plastic tubing. The transparent polyvinyl chloride or other suitable plastic tubing shall be flexible and pliable. The tubing shall be free from all liquid lubricants and constitutents which oxidize or exude to the surface of the tubing.

3.1.2 Adapters. Adapters shall be fabricated from rigid plastics.

3.1.3 Control clamps. The control clamps shall be fabricated from a rigid plastic such as polypropylene or nylon.

3.2 Style, design and dimensions. The unit shall be a plastic coiled tube suitable for the transfusion of blood. There shall be permanently affixed a male Luer adapter at one end and a female Luer adapter at the other end of the coiled tube. Each adapter shall be provided with a readily removable, plastic end protector. There shall be a Y-joint about 6 inches away from the male Luer adapter. The free arm of the Y-joint shall have a resealable rubber stopper. A control clamp shall be present. The unit shall be such that blood can be warmed from 35° F to 98° F prior to transfusion (see 6.4).

3.2.1 Tubing. The tubing shall be fabricated of material as indicated in 3.1.1. It shall have a length of 26-1/2 $\pm$ 3-1/2 feet, be precoiled, have an inside diameter of 0.120 $\pm$ .005 inch and a nominal outside diameter of 0.170 inch. It shall have a fluid capacity of approximately 58 $\pm$ 9 ml. Loops of the coil shall adhere firmly to one another and the coil shall have an inside diameter of approximately 4 inches. The coil shall be self-sustaining in an ovoid or circular shape without extrinsic support. The tubing shall be capable of being stripped from the coil without tearing the tubing sidewall.

3.2.2 Adapters. The adapters shall be permanently attached to the ends of the plastic tubing in a leakproof manner.

Source: https://assist.dla.mil -- Downloaded: 202610-21T09:01Z
Check the source to verify that this is the currentversion before use.

3.2.2.1 Male adapter. Male adapter shall be permanently affixed to the tubing, approximately 1-1/2 inches in overall length, and fabricated of rigid plastic. The taper shall comply with the requirements of ANSI Standard Z70.1 and shall form a tight, leak-proof fit with a female Luer taper in accordance with 4.4.8.

3.2.2.2 Female adapter. Female adapter shall be approximately 7/8 inch in overall length and shall make a tight, leak-proof fit with a male Luer needle adapter conforming with the requirements of ANSI Standard Z70.1 (see 3.7). The adapter shall be permanently connected to the tubing.

3.2.2.3 End protectors. End protectors shall be fabricated from a transparent or translucent plastic. End protectors shall be tubes or sheaths constructed in such a manner that when placed over the open ends of the coiled tube shall, after sterilization, maintain sterility, freedom from pyrogens, and integrity of the entire fluid path until manually removed. End protectors shall be readily removable. The female adapter end protector may contain a cotton microbiological barrier which acts as an easy entry for cold sterilization media.

3.2.3 Control clamp. The clamp if it is a screw-type clamp shall be one in which the screw shall be so designed as to make the screw removal difficult. The flow clamp shall provide a constant setting and quick shutoff from the operating position, shall be movable, and be capable of one-handed operation. To assure constant delivery of fluid, it shall be impossible for the clamp setting to be changed drastically by an accidental jerk on the tubing. When tested as specified in 4.4.3, the clamp shall be capable of accomplishing complete quick shut-off, no falling drop, from the male needle adapter. In addition, when the clamp is operated from its fully open position to complete shutoff, and this operation is repeated ten times at the same position on the tubing, the tubing shall not be cut, pierced, or damaged to cause leakage or restriction of flow of solution. Flow clamp or flow control shall be positioned between the coil and the Y-type injection site in the open position.

3.2.4 Y-type injection site. The "Y" injection site shall be fabricated from a rigid, clear, transparent plastic which meets the requirements specified in 3.1. An upper branch of the "Y" shall be connected to the coiled tube end, and the leg of the "Y" shall be connected to a piece of tubing approximately six inches long. The remaining upper branch shall be closed by a rubber stopper permanently and securely sealed to the branch.

3.2.4.1 Rubber stopper. Rubber stopper shall be fabricated from material as specified in 3.1. The stopper shall be capable of resealing subsequent to puncture with a non-siliconized 18 gage needle cannula. When tested for resealing as specified in 4.4.4, there shall be no falling drop within ten seconds.

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T09:01Z
Check the source to verify that this is the current version before use.

3.3 Manufacturer's record. The manufacturer shall place a lot control number on the paper label, or on the printed carton, which can be verified against the manufacturer's records of all tests made which shall be available for inspection by any duly qualified medical officer of the Armed Forces, National Institutes of Health, or any official of the Federal Food and Drug Administration. No shipment shall be made until all the tests have been completed, verified, and recorded in the permanent records of the manufacturer.

3.4 Sterilization, safety, and freedom from pyrogens. The blood warming coiled tube shall be safe (non-toxic), sterile in its fluid path, and pyrogen-free. All the above shall be in accordance with current revision of the minimum requirements of the National Institutes of Health for citrated whole blood. The fluid path and end connectors shall be pyrogen-free when tested as specified in 4.4.6 and sterile when tested with a biological indicator as specified in 4.4.5. The sets shall be safe when tested for safety as specified in 4.4.7.

3.5 Certificate of sterility. A certificate of sterility, stating that samples from each sterilizer lot have been tested and found to be sterile, shall be submitted to the procuring activity. The certificate of sterility shall include:

a. Item identification.
b. National stock number (NSN).
c. Contract or purchase order number.
d. Sterility lot or control number.
e. Date of sterilization.
f. Quantity of product in sterilizer lot.
g. Type of sterilization (gas, radiation, heat, steam, etc.).
h. Form of biological indicator (i.e., inoculated paper spore strip or carrier, inoculated product, or inoculated simulated product).
i. Test organism(s) and population density used.
j. Test medium, incubation temperature, incubation time, and dates of tests.
k. Number of test units.
l. Statement that samples comply with sterility tests.
m. Signature of head of laboratory or a designated qualified alternate.

When inspection is made at destination, the certificate of sterility shall be submitted by the contractor to the Directorate of Medical Materiel, ATTN: DPSC-ATQ, Defense Personnel Support Center, 2800 South 20th Street, Philadelphia, PA 19101, and a copy forwarded with each shipment to the consignee. When inspection is made at source, certificate of sterility shall be furnished to the cognizant Government Quality Assurance Representative for submission to the procuring activity.

3.5.1 Sterility lot. A sterility lot is that single quantity of product subjected to the same manufacturing operation and simultaneously sterilized in the same sterilizer chamber. Each unit package shall be clearly marked with a sterility lot (control) number and date of sterilization as specified in 5.3.1. Each sterility lot shall be traceable to the pertinent manufacturing lot(s).

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T09:01Z
Check the source to verify that this is the current version before use.

3.5.2 Certificate of freedom from pyrogens. A certificate of freedom from pyrogens, listing all lot or control numbers, shall state that samples from each lot or control have been tested for pyrogens as specified in 3.4 and were found to comply with such requirements. The certificate shall reference the method used in determining freedom from pyrogens.

3.5.3 Certificate of safety. A certificate of safety, listing all lot or control numbers, shall state that samples from each lot or control have been tested for safety as specified in 3.4, and were found to comply with such requirements. The certificate shall reference the method used in determining safety.

3.6 Leakage and separation. Coiled tubes shall be tested for leakage and separation of set as specified in 4.4.8. Sets which show leakage of one or more drops or separation at any part of the set shall be rejected.

3.7 Fluid path. Fluid path shall be clear and shall be capable of delivering one liter of liquid in less than 16 minutes when tested as specified in 4.4.9.

3.8 Security of assembly. Blood warming coils shall have the adapters and Y-joint bonds tested as specified in 4.4.10. The assemblies shall withstand a load of 4 pounds for 15 minutes.

3.9 Instruction. A detailed instruction sheet for proper use of the coiled tubes shall be provided with each unit package.

3.10 Workmanship. Workmanship shall be first class throughout. The coiled tubes shall be free from defects which detract from their appearance or impair their serviceability.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T09:01Z
Check the source to verify that this is the current version before use.

4.1.1 Records. Records of examinations and tests performed by or for the contractor shall be maintained by the contractor and made available to the Government, upon the Government's request, at any time, or from time to time, during the performance of the contract and for a period of 3 years after delivery of the supplies to which such records relate.

4.1.2 Inspection. Inspection, as used in this specification, is defined as both examination (such as visual and auditory investigation without the use of special laboratory appliances or procedures) and testing (determination by technical means of physical and chemical properties) of the item.

4.1.3 Certificates of quality. Certificates of quality supplied by the manufacturer(s) of the plastic components may be furnished in lieu of actual performance of such testing by the contractor, provided lot identity has been maintained and can be demonstrated to the Government. The certificate shall include the name of the contractor, the contract number, the name of the manufacturer or supplier, the NSN, the Item Identification, the name of the component/material, the lot number, the lot size, the sample size, the date of testing, the test method, individual test results, and the specification requirements.

4.2 Sampling.

4.2.1 For examination. Sampling for examination shall be conducted in accordance with MIL-STD-105 and as indicated in table I. The unit of product for examination purposes shall be one blood warming coil complete with accessories.

TABLE I. Sampling criteria for examination.

Type of defect	Inspection level	AQL (percent dective)
Major	I	1.0
Minor	I	2.5

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T09:01Z
Check the source to verify that this is the current version before use.

4.2.2 Sampling for test. Sampling for tests shall be conducted in accordance with MIL-STD-105 and as indicated in table II. The unit of product for test purposes shall be one blood warming coil complete with accessories. Where possible, the same sample shall be used for the determination of two or more test characteristics.

TABLE II. Sampling criteria for tests.

Characteristic	Requirement	Test procedure	Inspection level	AQL (percent defective)
Class II plastic	3.1	4.4.1	S-1 1/	1.5
Lead	3.1	4.4.2	S-1	1.5
Flow control	3.2.3	4.4.3	S-1	1.5
Rubber stopper	3.2.4.1	4.4.4	S-2	1.0
Sterility	3.4	4.4.5	S-2 1/	1.0
Non-pyrogenicity	3.4	4.4.6	S-2 1/	1.0
Safety	3.4	4.4.7	S-2 1/	1.0
Leakage and separation	3.6	4.4.8	S-2	1.0
Fluid path	3.7	4.4.9	S-1	1.5
Security of assembly	3.8	4.4.10	S-2	1.0

1/ Inspection level shall be the higher of Table II or USP.

4.3 Examination. The blood warming coil shall be examined to determine compliance with all the requirements contained in this specification. Examination shall be conducted as specified in tables I and III. Examination, however, shall not be restricted to defects listed in table III. Any blood warming coil in the sample containing one or more defects shall be rejected, and if the number of defective blood warming coils in any sample exceeds the acceptance number for that sample, the lot represented by that sample shall be rejected.

TABLE III. Classification of defects.

Categories	Defects
Major	
101	Luer adapters not of correct dimensions.
102	Crack in an adapter.
103	Flow regulator is too large to center tube under screw.
104	Flow regulator is too small to permit full flow through tube.
105	Presence of foreign materials in fluid path.
106	Fluid volume capacity is not approximately 50 ml.
Minor	
201	Burrs on exterior of adapters.
202	Instructions missing, incomplete, or illegible.
203	Tubing not clear or free from discoloration.
204	Presence of foreign materials but not in fluid path.

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T09:01Z
Check the source to verify that this is the current version before use.

4.3.1 Packaging inspection. The inspection of the unit package, intermediate package, packing and marking for shipment and storage shall be in accordance with the quality assurance provisions of the applicable container specification and the marking requirements of MIL-STD-129.

4.4 Tests. Tests shall be conducted to determine compliance with specification requirements.

4.4.1 Class II plastic. Testing shall be accomplished in accordance with Class II (70° C) of USP.

4.4.2 Test for lead. Ten grams of sample shall be taken, ignited to char, and then further ignited in a muffler at 700° C. to a white ash. The ash shall be boiled in 15 ml of 10 percent nitric acid, filtered into a volumetric flask and made up to 100 ml. An aliquot portion, equivalent to one gram of sample, shall be taken and placed in a separator. The dithizone test for lead, as given in the current edition of the Pharmacopeia of the United States, shall be applied using a control of 5 ppm or 12 ppm of lead solution as given in the USP test. If the sample solution is darker than the limit, smaller aliquots shall be used until the color of the control is matched by the sample solution and the concentration of lead in ppm is determined.

4.4.3 Flow control. The finished tube shall be connected, in the normal manner, to a standard solution container filled with saline solution or water. The container, with tube connected thereto, shall be suspended, in the normal manner, at a sufficient height to permit two feet of the tubing on each side of the coil to hang vertically. The ability of the clamp to accomplish complete shutoff with a constant head of liquid shall be tested with the clamp at a point on the plastic tubing within two inches above the "Y" component. The tests shall be performed with the tubing hanging vertically at all times.

4.4.4 Reseal test (rubber stopper). This test shall be applied to those samples previously subjected to the leakage test (4.4.8). The blood warming coil shall be connected to a full, suitably vented, one liter bottle of water, saline solution, or dextrose solution. The bottle shall be hung in a normal manner so that the liquid level in the bottle shall be three feet above the injection site. The coil shall be completely filled with liquid and sealed at the male needle adapter. The "Y" shall be placed in a horizontal position and shall be punctured thru the stopper with an 18 gage non-siliconized needle and the needle immediately withdrawn.

Source: https://assist.dla.mil -- Downloaded: 202610-21T09:01Z
Check the source to verify that this is the current version before use.

4.4.5 Sterility test. The fluid path of the blood warming coil shall be tested for sterility in accordance with the method specified in the USP for parenteral products using inoculated product, inoculated simulated product, or paper spore strip (carrier). Product containing a biological indicator shall be marked or labeled so that it clearly differentiates from product intended for distribution. The incubation medium and minimum time in each case shall be soybean casein digest for 7 days for inoculated product, inoculated simulated product, or paper spore strip (carrier). The incubation temperature shall be as indicated in paragraph 4.4.5.1. The minimum number of test units, regardless of lot size, shall be 10, except for inoculated carrier which shall be 20 for Type I Sterilization and 30 for Type II Sterilization including inoculated carrier and product test units. The inspection level shall be as indicated in table II, Sampling criteria for tests.

4.4.5.1 Inoculated product or simulated product. For each type of sterilization, the organism used shall be as listed below:

Type of Sterilization	Organism	Incubation Temperature
Gas	B. subtilis var. niger (or globigii)	30° - 35°C
Radiation	B. pumilus	30° - 35°C

The number of viable cells shall be sufficient to provide assurance that the product has been sterilized.

4.4.5.2 Inoculated carrier. Inoculated carriers (paper spore strips) if used shall be in accordance with MIL-S-36586. Other paper carriers shall be inoculated to be of equivalent resistance. The method of incubation and number of test units shall be as indicated in 4.4.5 Sterility test and 4.4.5.1 Inoculated product or simulated product. The method of sterilization will control the one organism in the spore strip to be tested for as shown in 4.4.5.1. NOTE: When an inoculated carrier is chosen, product samples must also be tested as required by the USP.

4.4.5.3 Alternate method. Suppliers conducting sterility tests or designating lots in a manner other than that indicated herein shall submit proof that the alternate method provides equal assurance concerning the sterility of the product. However, all sterility tests shall be conducted using biological indicators. Requests for approval of any alternate method of testing for sterility shall be made to the Contracting Officer, and approval obtained before employment of the alternate method on a government contract.

Source: https://assist.dla.mil -- Downloaded: 202610-21T09:01Z
Check the source to verify that this is the current version before use.

4.4.5.4 Product testing. Testing for sterility of the product after receipt by the government shall be without a biological indicator and in accordance with the USP.

4.4.6 Freedom from pyrogens. Samples of the blood warming coil shall be tested for freedom from pyrogens in accordance with the USP for Transfusion and Infusion Assemblies.

4.4.7 Safety (non-toxicity). The blood warming coils shall be tested for safety in accordance with the USP for Transfusion and Infusion Assemblies.

4.4.8 Leakage and separation of set. The female Luer adapter of the blood warming coil shall be attached to a plug gage of the type shown in ANSI Z70.1 (except that the gage shall have been provided with a fluid path) and a water pressure source. The water is allowed to flow and displace all air. The blood warming coil shall be conditioned by immersion for 15 minutes prior to the testing and held during the testing in a water bath at 97° F $\pm$ 3° F. A liquid pressure gage with a standard Luer hub which has been drilled shall be attached to the male Luer adapter. The pressure shall be increased to 45 psig. This pressure shall be maintained for 30 minutes. There shall be no leakage or separation.

4.4.8.1 Alternate test. As an alternate test, 10 psig air pressure shall be established in the coil and the coil immersed for 10 seconds in water. The appearance of one bubble shall constitute a failure.

4.4.9 Fluid flow test. The blood warming coil shall be connected to a full, suitably vented, one liter bottle of water, saline or dextrose solution. The bottle shall be hung in the normal manner so that the liquid level is three feet above the male needle adapter. The liquid shall be allowed to fill the entire fluid path. The time necessary to empty completely the one liter bottle shall be determined.

4.4.10 Security of assembly. The finished blood warming coil shall be tested for separation of each component to the tubing by securing each component or tubing in the vertical position and suspending a four-pound weight from the lower element for 15 minutes. There shall be no separation.

5. PACKAGING

5.1 Packaging. Packaging shall be level A or C as specified (see 6.2).

5.1.1 Unit of issue. Each blood warming coil, as specified, constitutes one unit of issue.

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T09:01Z
Check the source to verify that this is the current version before use.

5.1.2 Level A.

5.1.2.1 Unit packaging. Each blood warming coil shall be packaged in a box conforming to PPP-B-566 or PPP-B-676. Closure shall be as specified in the appendix of the applicable box specification.

5.1.2.2 Intermediate package. Six unit packages shall be packed in a 4 mil plastic bag and the bag closed by heat sealing or by folding and securing with pressure sensitive tape or by using a twist tie. The sealed bag shall be further packaged in a box conforming to PPP-B-636, class domestic. Closure shall be as specified in the appendix of the box specification.

5.1.3 Level C. Each blood warming coil shall be packaged in a manner that will afford adequate protection against deterioration and physical damage during shipment from the supply source to the first receiving activity.

5.2 Packing. Packing shall be level A, B or C as specified (see 6.2).

5.2.1 Level A. Twelve intermediate packages (72 units) shall be packed in an exterior container designed for a type 2 load and conforming to PPP-B-636, class weather-resistant. Closure, strapping and waterproofing shall be as specified in the appendix of the box specification.

5.2.2 Level B. Twelve intermediate packages shall be packed in an exterior container designed for a type 2 load and conforming to PPP-B-636, class domestic. Closure shall be as specified in the appendix of the box specification.

5.2.3 Level C. The packaged blood warming coils shall be packed in shipping containers in a manner that will afford adequate protection against damage during direct shipment from the supply source to the first receiving activity. These packs shall conform to the applicable carrier's rules and regulations.

5.3 Marking.

5.3.1 Unit package. Each unit package shall be marked in accordance with MIL-STD-129. The sterility lot (control) number, contract or purchase order number, name of contractor and method of sterilization shall also be shown. The method of sterilization (G-Gas, R-Radiation, Sterilization) shall be shown following the level of packaging and date of packaging. Example: "A-2/77-G." Final sterilization shall be on or after indicated packaging date. In addition, information required by the Federal Food, Drug and Cosmetic Act, and detailed instructions as specified in 3.9 shall be printed on the container, or included as an insert therein. Marking shall include the following legends:

"DO NOT IMMERSE IN WATER OVER 105° F."
"STERILE, NON PYROGENIC"
"DO NOT RE-USE"
"FLUID PATH AND AREAS UNDER PROTECTIVE COVERING OF THE TUBE ARE STERILE IF TUBE COVERS ARE IN PLACE" or similar statement.

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T09:01Z
Check the source to verify that this is the current version before use.

5.3.2 Intermediate package. Each intermediate package shall be marked in accordance with MIL-STD-129. When labels are utilized, waterproofing shall be required only when applicable carton is fabricated of water-resistant material. In addition, marking shall include the method of sterilization, markings as specified in 5.3.1. The sterility lot (control) number, contract or purchase order number, and the name of contractor shall be shown.

5.3.3 Exterior container. Exterior container shall be marked in accordance with MIL-STD-129. In addition, marking shall include the method of sterilization markings as specified in 5.3.1, and the sterility lot (control) number.

Note: Each lot number shall be traceable to a specific sterilized charge.

6. NOTES

6.1 Intended use. The blood warming coil is intended to warm blood from a blood bank prior to transfusion in order to prevent hypothermia.

6.2 Ordering data. Procurement documents should specify the following:

(a) Title, number and date of this specification.
(b) Applicable levels of packaging and packing (see 5.1 and 5.2).
(c) National stock number.

6.3 This specification covers the following item appearing in the Federal Supply Catalog:

National Stock No.	Item Identification
6515-00-926-2205	TUBE, COILED, BLOOD WARMING, Disposable, Plastic

The TUBE, COILED, BLOOD WARMING is intended to be used in conjunction with NSN's 6515-00-089-3470, 6515-00-558-1509 and 6515-00-958-7795 BLOOD RECIPIENT SETS, INDIRECT TRANSFUSION.

6.4 Heat exchange. No test method has been included in this specification for heat exchange capability inasmuch as the tube length, wall thickness, and inside diameter are specified in section 3. These points of control plus an adequate constant temperature bath should be a proper means to warm the blood prior to transfusion.

6.5 Changes. Asterisks to indicate changes have been omitted in this specification due to the extensiveness of the changes.

Custodians:
Army - MD
Navy - MS
Air Force - 03

Preparing activity:
Defense Logistics Agency - DM

Project No. 6515-2076

Source: https://assist.dla.mil -- Downloaded: 202610-21T09:01Z
Check the source to verify that this is the current version before use.

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T09:01Z
Check the source to verify that this is the current version before use.

STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

OMB Approval
No. 22-R255

INSTRUCTIONS: The purpose of this form is to solicit beneficial comments which will help achieve procurement of suitable products at reasonable cost and minimum delay, or will otherwise enhance use of the document. DoD contractors, government activities, or manufacturers/vendors who are prospective suppliers of the product are invited to submit comments to the government. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements. Attach any pertinent data which may be of use in improving this document. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity.

DOCUMENT IDENTIFIER AND TITLE
MIL-T-36682A TUBE, COILED, BLOOD WARMING

NAME OF ORGANIZATION AND ADDRESS

CONTRACT NUMBER

MATERIAL PROCURED UNDER A
☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?
   A. GIVE PARAGRAPH NUMBER AND WORDING.

   B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY DOCUMENT REQUIREMENT CONSIDERED TOO RIGID

3. IS THE DOCUMENT RESTRICTIVE?
   ☐ YES ☐ NO *(If "Yes", in what way?)*

4. REMARKS

SUBMITTED BY *(Printed or typed name and address - Optional)*

TELEPHONE NO.

DATE

DD FORM 1 JAN 72 1426

REPLACES EDITION OF 1 JAN 66 WHICH MAY BE USED

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T09:01Z
Check the source to verify that this is the current version before use.

FOLD

DEFENSE PERSONNEL SUPPORT CENTER
2800 South 20th Street
Philadelphia, PA 19101

POSTAGE AND FEES PAID
DEFENSE SUPPLY AGENCY
DoD - 304

OFFICIAL BUSINESS
PENALTY FOR PRIVATE USE $300

HEADQUARTERS, DEFENSE PERSONNEL SUPPORT CENTER
ATTN: DIRECTORATE OF MEDICAL MATERIEL, CODE ATT
2800 SOUTH 20TH STREET
PHILADELPHIA, PA 19101

FOLD

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T09:01Z
Check the source to verify that this is the current version before use.